



DIGITAL LOGIC PROJECT

Photostat machine system design

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Course: SECR1013

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1. Dedication and Acknowledgement

This project is dedicated to our Lecturer, Dr. Nur Haliza. We would like to express our gratitude towards her, for teaching us Digital logic this semester, and equipping us with the knowledge needed to complete this project. Not only did we have the opportunity to learn about digital circuits and components, but also its real-life applications.

We would also like to express our thanks to our group members, who worked hard to complete this project to present it in the best way possible. Working on this project truly inspired us and improved our thinking skills and problem-solving skills, along with our ability and interact and cooperate in a team.

2. The background

In this project, we will implement a password-protected photostat machine, which allows to use to input copying type and size and a 4-bit number for the requested number of copies. The full circuit is drawn using dees and includes combinational circuit and sequential circuits.

Combinational circuit components include:

- 4-bit comparator
- multiplexer
- demultiplexer
- 4 bit decoders
- basic gates
- switches
- LEDs
- Seven segment display

Sequential circuit components include:

- 4 bit up Counter
- Clock disabler

Enhanced features:

- 4-digit password

- copying type

-copying size

3. problem and solution

Unauthorized use can often be a problem, so to avoid this, we have implemented a password using 4, 4-bit decoders to allow the user to input a four-digit password. Firstly, the user has to turn on the power button then input the password (Which is already set). If the password entered is wrong or the photostat machine is turned off, the user may not select copying options or start photocopying. If the password is correct, the LEDs light up, and the user may proceed. Our photocopying machine allows users to select the copying type and size and displays the option chosen using a seven-segment display. The user selects the options, then inputs the number of copies and presses clear to start the copying process. The number of requested copies and copies produced is displayed. The led connected to the clock disabler turns off and on, indicating that it is processing. The 4-bit counter counts up and is connected to one of the 7 segment displays. The comparator keeps comparing this value with the value input by the user until it becomes the same. After that, the LEDs turn off. The clock disabler stops the counting, indicating that the copying process is over. The user may press clear (to make it 0) and select other options to restart the copying.

The complete process is shown in the Block diagram below.

4. Block diagram

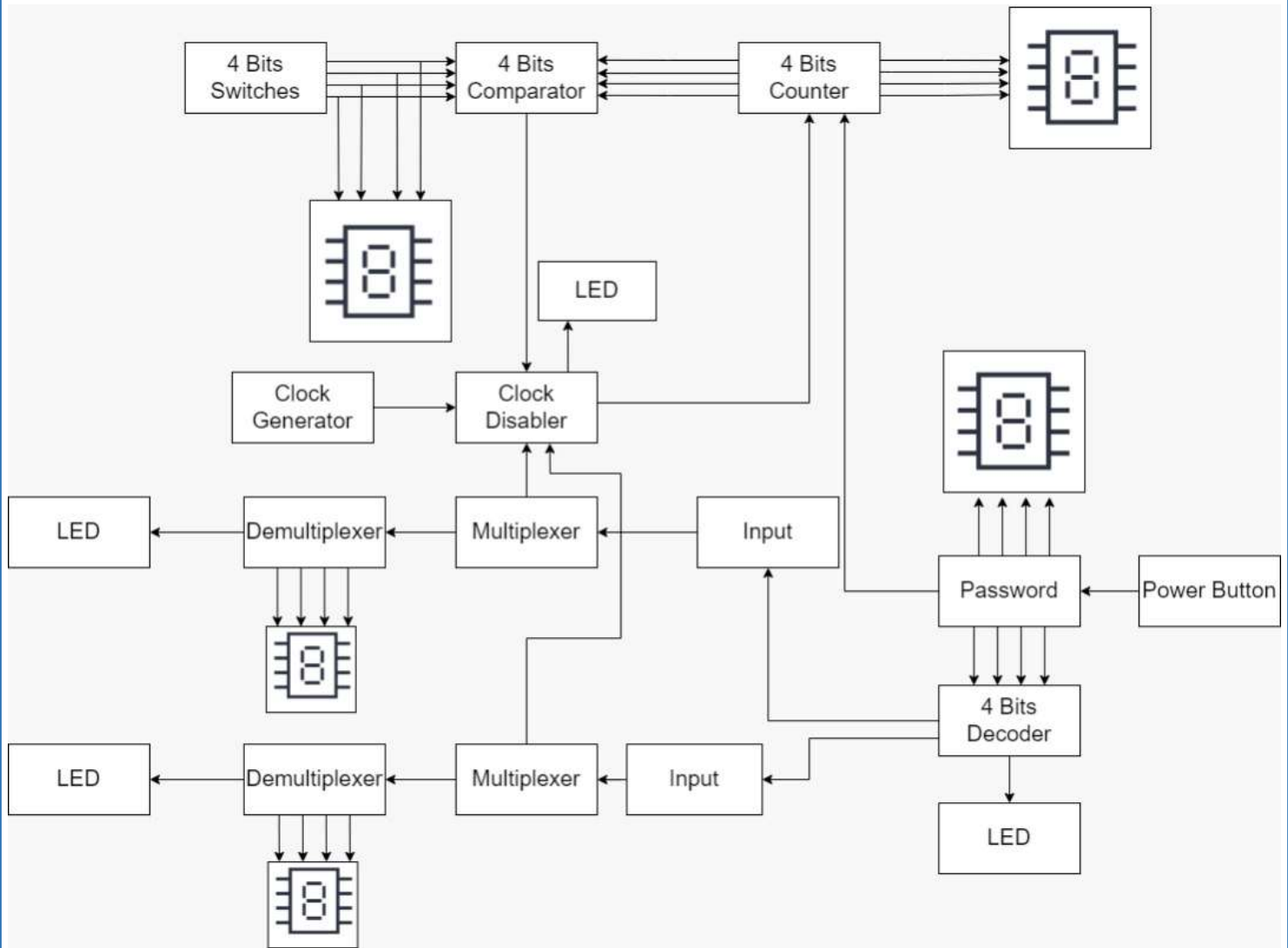


Fig 1: Block diagram

5. The requirements

Input switch

The user can enter the quantity of copies they wish to print using four input switches.

Multiplexer

For each multiplexer, there are two user inputs. The user can choose the photocopying properties such as paper size and color. These multiplexers carry out the function for choosing properties.

Decoder

The 4-bit decoder is used to decode the 4 digit binary password. The user provides the password with the help input switches and the power button should be on for the decoder to work.

Demultiplexer

There are two demultiplexers, and each is connected to the multiplexer's output since this component is required to determine the properties-type and properties-size that the user has chosen.

Clock disabler

The clock disabler is active only when all four inputs are high, and it is used to start and stop the counter operation when the number of copies and printed copies is reached.

4-bit count up counter

The counter begins and ends counting depending on the clock enabler and the password entered by the user.

7-Segment Display

The 7-segment display shows the password output, the user's selection of properties-type and properties-size, and the number of copies requested and created.

4 bit-Comparator

The function of the 4-bit comparator is to compare the value of requested copies for photocopy and actually printed copies.

Power button

The power button input switch allows the user to choose whether to turn on or off the printer. The decoder will only work if the power button is active.

LED

Each decoder's output is coupled to two LEDs. When the user selects the input value, it will light.

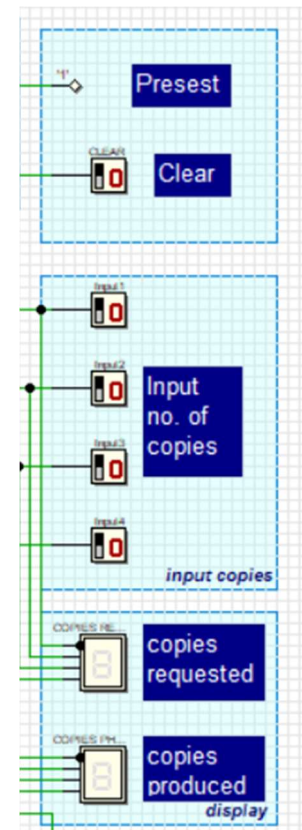
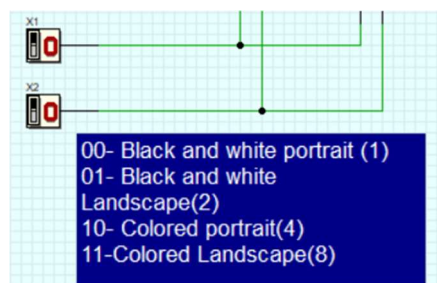
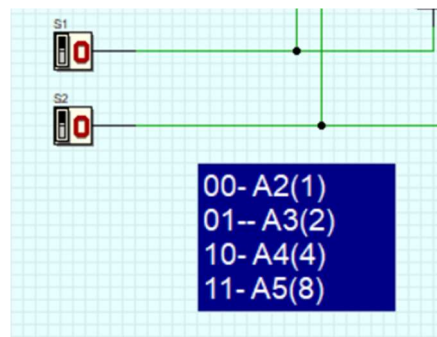
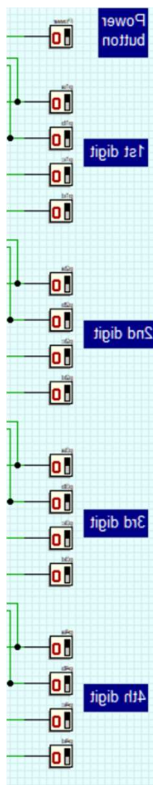
Clear button

When switched on (1) starts copying and when switched off (0) allows user to restart copying (select other options and use the machine again)

6. System implementation

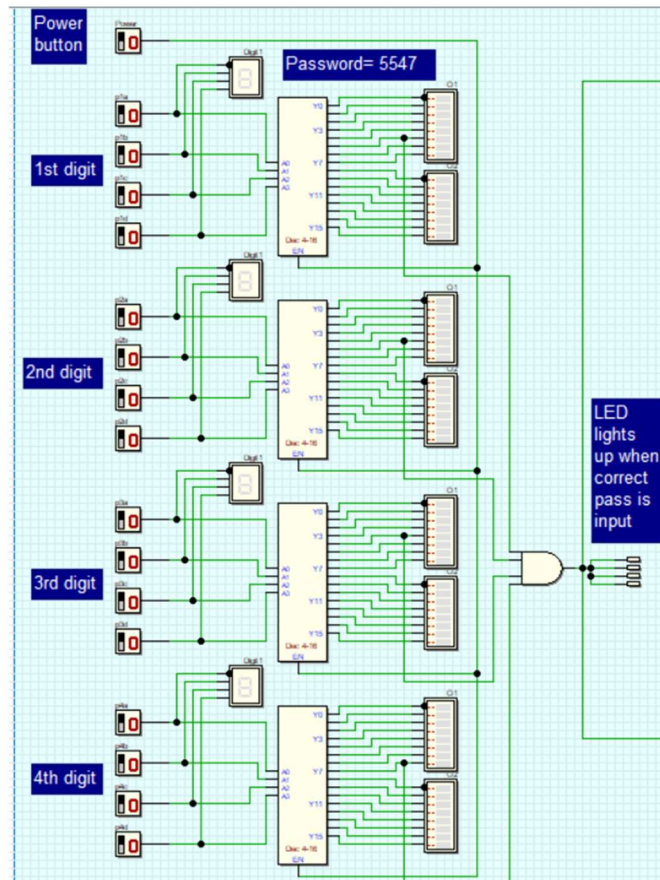
I. Input switch

There are 4 input switches for the user to input the number of copies they want to print. Each of the switches represents a single bit respectively, which are Input 1, Input 2, Input 3 and Input 4. Input 1 represents LSB, and Input 3 represents MSB. Hence, the user can input values ranging from 0 to 15(F). The output of the input switch is connected 7-segment display to display the number of copies requested and produced and the comparator. There are also 16 input switches for the user to input the 4-digit password connected to the 4-bit decoder to decode the set of binary passwords. There is also has one input switch for activating the clear button. When the clear button is set to be 1, it will print by following properties-type and properties-size. It will stop printing when the number of copies is the same as the requested number. Additionally, there are two input switches for choosing the copy type and two for determining the size. Lastly, there is an input switch used as a power button to start and stop the photocopying machine.



II. Decoder

We want to set a 4-digit password. So, we use four 4-bit decoders to decode the set of binary passwords. Each decoder represents each digit of the password. There is a power button input switch to select either turn on or turn off the printer. The enable pin of the 4-bit decoder is connected to the power button. It means that the decoder will only activate when the power button is activated. The output of each decoder is connected to two LED arrays named Q1 and Q2. It will light up when the user has selected the input value ranging from 0 to F for each decoder. The output of the decoders is connected to an AND gate, and the AND gate is connected to the LED. Once the LED light up, it means the correct password is input by the user, and the printer power is on, and it will allow the user to work with this photocopy machine. The output of the decoder is also connected to AND gate with the multiplexer and flip-flop, which means activating the properties-type, properties-size, and Asynchronous up counter after the password had to be correct. All the AND gate is connected to the clock enabler. When all the outputs from the decoders are high, it will activate the clock enabler.

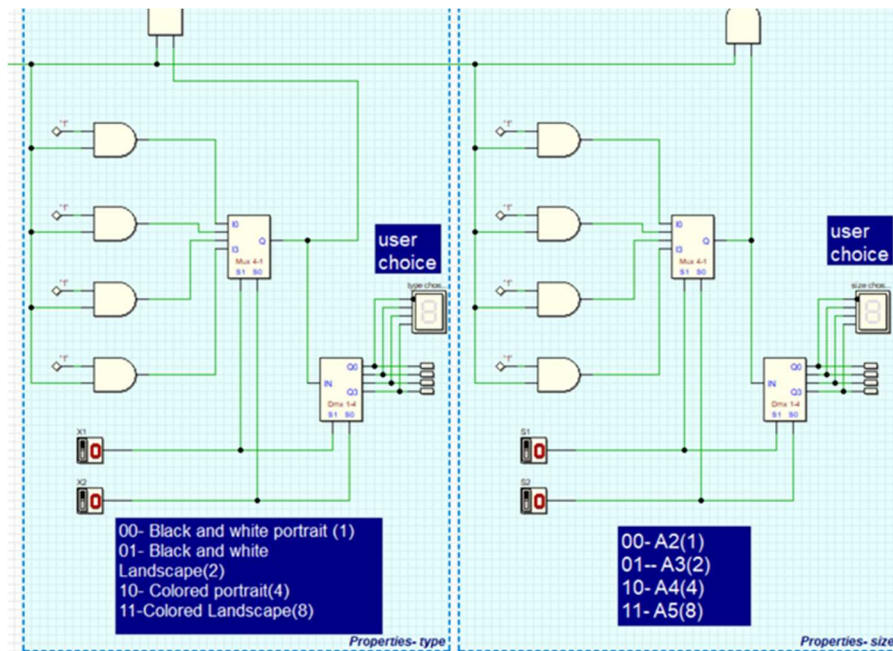


III. Multiplexer

There are 2 multiplexers for the user to select the printing properties that the user wants to print. There have two user inputs for each multiplexer. The two user inputs for the first multiplexer are called X0 and X1; The two user inputs for the second multiplexer are called S0 and S1. Those inputs are used to choose the properties that the user wants to print in each multiplexer. For the first multiplexer, both X0 and X1 are 0, the user will select the black and white with portrait printing; While X1=0 and X0=1, the user will select the black and white with landscape printing; While X1=1 and X0=0, the user will select the coloured with portrait printing; While both X1 and X0 is 1, the user will select the coloured with landscape printing. For the second multiplexer, both S0 and S1 are 0. The user will choose the A2 paper printing; While S1=0 and S0=1, the user will select the A3 paper printing; While S1=1 and S0=0, the user will select the A4 paper printing; Both S1 and S0 are 1, the user will select the A5 paper printing. Both properties-type and properties-size are connected to the 4-bit decoder using AND gate. If the user enters an invalid password, it will not proceed to this step.

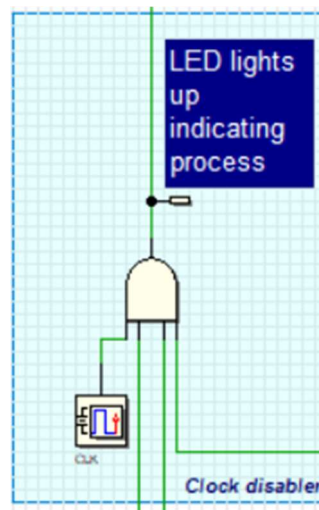
IV. Demultiplexer

There are 2 demultiplexers that are connected to the output of the first and second multiplexers. Each demultiplexer is connected to the output of the multiplexer because this component is used to determine the properties-type and properties-size that the user had selected. The output of both multiplexers is connected to the LED and 7-segment display. Once the user is finished selecting the properties included properties-type and properties-size, the LED will light up by following the number of user choices which is 1, 2, 4 and 8. 7-segment display will display the number of user choices which is 1, 2, 4 and 8 to let the user for choosing the properties-type and properties size.



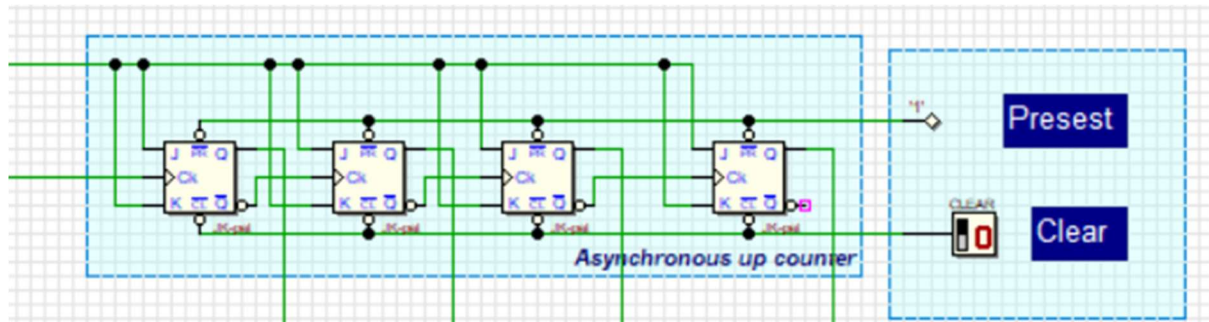
V. Clock Disabler

The clock Disabler is set up by using a 4-input AND gate. The input of the 4-input AND gate is the clock source, a signal sent by the properties-type, a signal sent by the properties-size, and a signal sent by the comparator. The clock disabler will only be active when all four inputs are high. The output of the 4-input AND gate is connected to the clock of the flip-flop of the counter. It used to start and stop the operation of the counter when the number of copies and printed copies had been met.



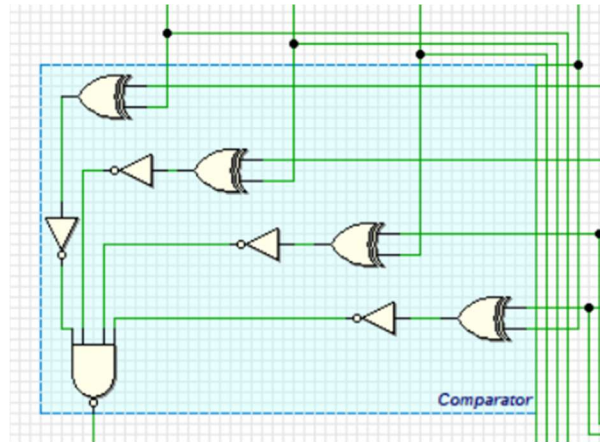
VI. 4-bit count up counter

The 4-bit counter that we use is 4-bit JK positive edge count-up counter. The counter starts or stops to count based on the clock enabler. It will start counting if the J and K inputs are connected to high input. While J and K inputs are connected to low input, it will stop. When the user input the correct password, the AND gate in the password site will implement the active high input. The counter will not start to count if the password is entered wrong. It will stop when the clock pulse is no longer received or reached the number of counts initialised by the user.



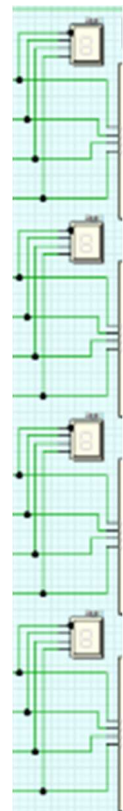
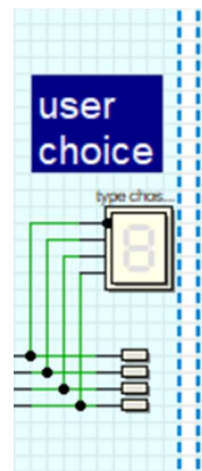
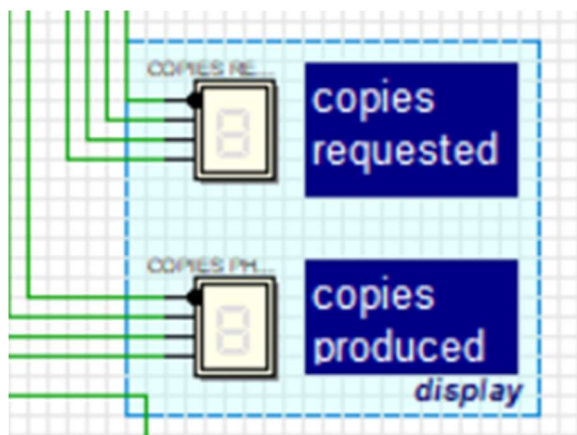
VII. 4 bit-Comparator

The comparator is set up by using four NOT gate and four 2-input XOR gates which are used to compare value from 2 sources, which are the input switches which are used to input the number of copies by the user and the 4-bit count up counter, which is used to count the number of copies that had been produced. The first XOR gate compares the least significant bit (LSB) of the 2 sources. If the number of copies requested and the number of copies produced are the same, the output will be 0, and it is sent to the NOT gate and converted into 1. The same principle can be applied for the second, the third and the fourth XOR gate. Then, the signal from all four gates will be sent to a NAND gate to convert it into the opposite signal. Therefore, when the input of the NAND gate is received the high input (1) given from input switches and the counter, it will send an output 0 to the clock enabler to stop the photocopying machine.



VIII. 7-Segment Display

The 7-Segment display displays the number based on the BCD value by converting the BCD code received into decimal numbers and letters. This component will display the decimal numbers 0 to 9 and letters from A to F by turning up the proper light of the 7-segment display. The password output, the user choice of properties-type and properties-size, and the number of copies requested and produced are connected to a 7-segment display. Since our number of copies produced and requested, and password number ranging from 0 to 15, this component will display the number ranging from 0 to F. The user choice will display the number, the number of user choices for choosing the properties-type and properties-size. (1, 2, 4 and 8)..



7. Full Deeds circuit

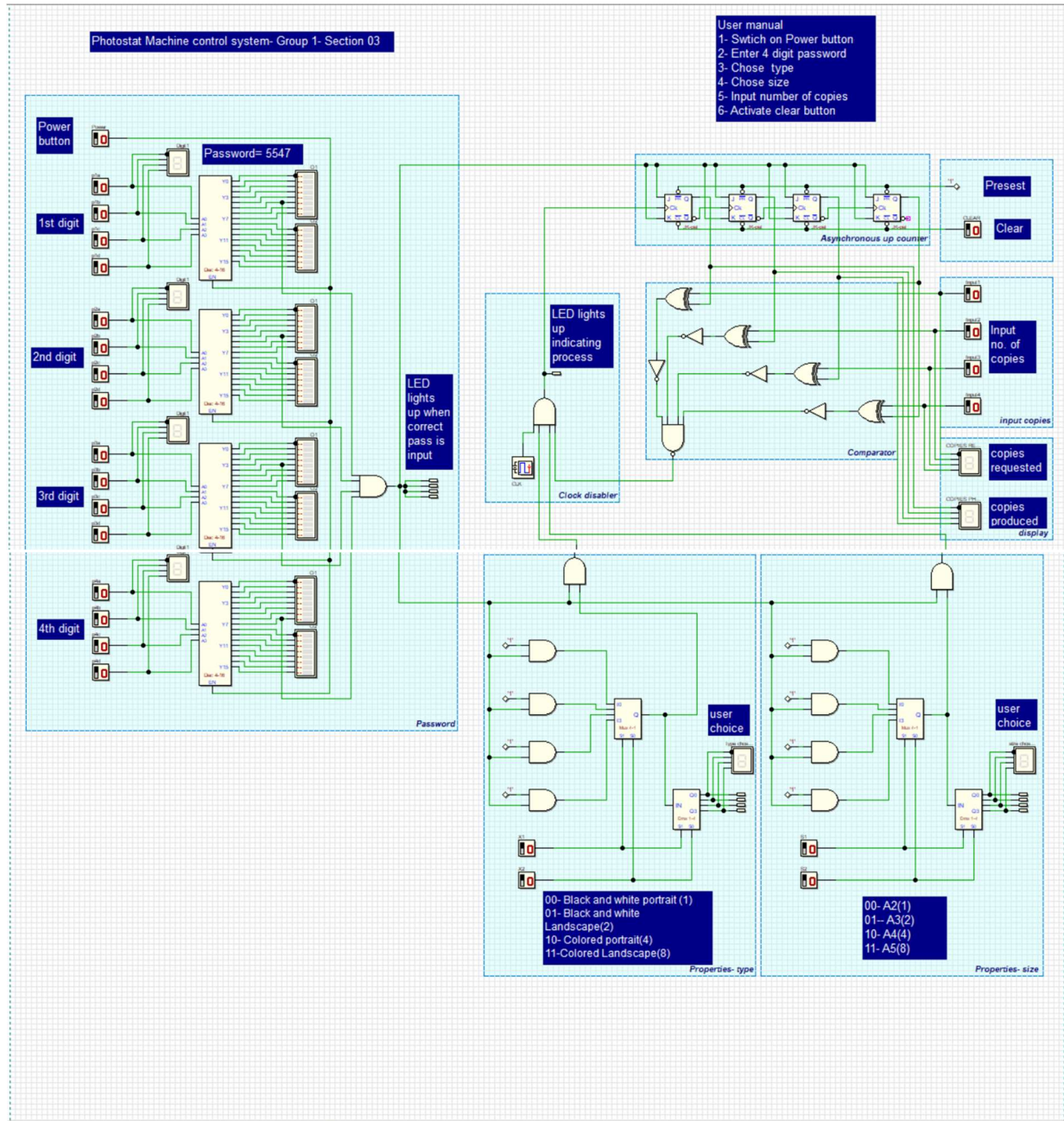


Fig 2: Full deeds circuit

8. Conclusion

To conclude, the project has a simple, user-friendly design. The circuit mainly consists of the 4-bit decoders, the input switches, seven segment displays and LEDs for outputs, a 4-bit comparator, 4-bit up counter and a multiplexer and demultiplexer. The password for security was a great addition to our circuit and was connected so that none of the other components works when it is wrong. Another great option we implemented was the properties. One of the issues we encountered was including an adder to output the price of copying. In future projects, we are hoping to implement even more complex circuits with more exciting options, and we are looking forward to using our knowledge in related courses and applying it to real-life situations

9. References

[1] Digital logic Module

[2] Digital logic lab

10. Appendices

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[3] Block diagram

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[4] Requirements

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[5] System Implementation

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[6] Deeds circuit

- Sondos Tamer Moustafa, A21EC0277

[7] conclusion

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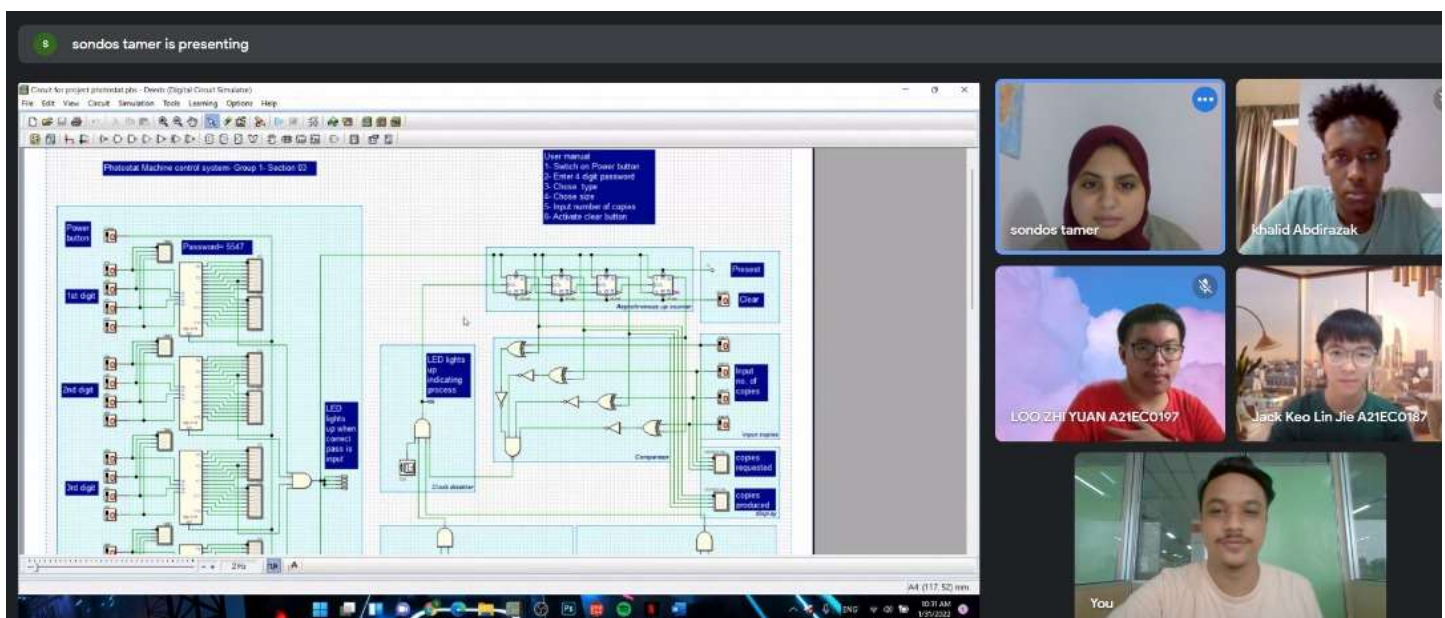


Fig 3: Group 1 working together on google meet

Video link:

https://drive.google.com/file/d/13WcRC_vQHSh4QgHqCzCnbKEKWUwmFqY1/view?usp=sharing